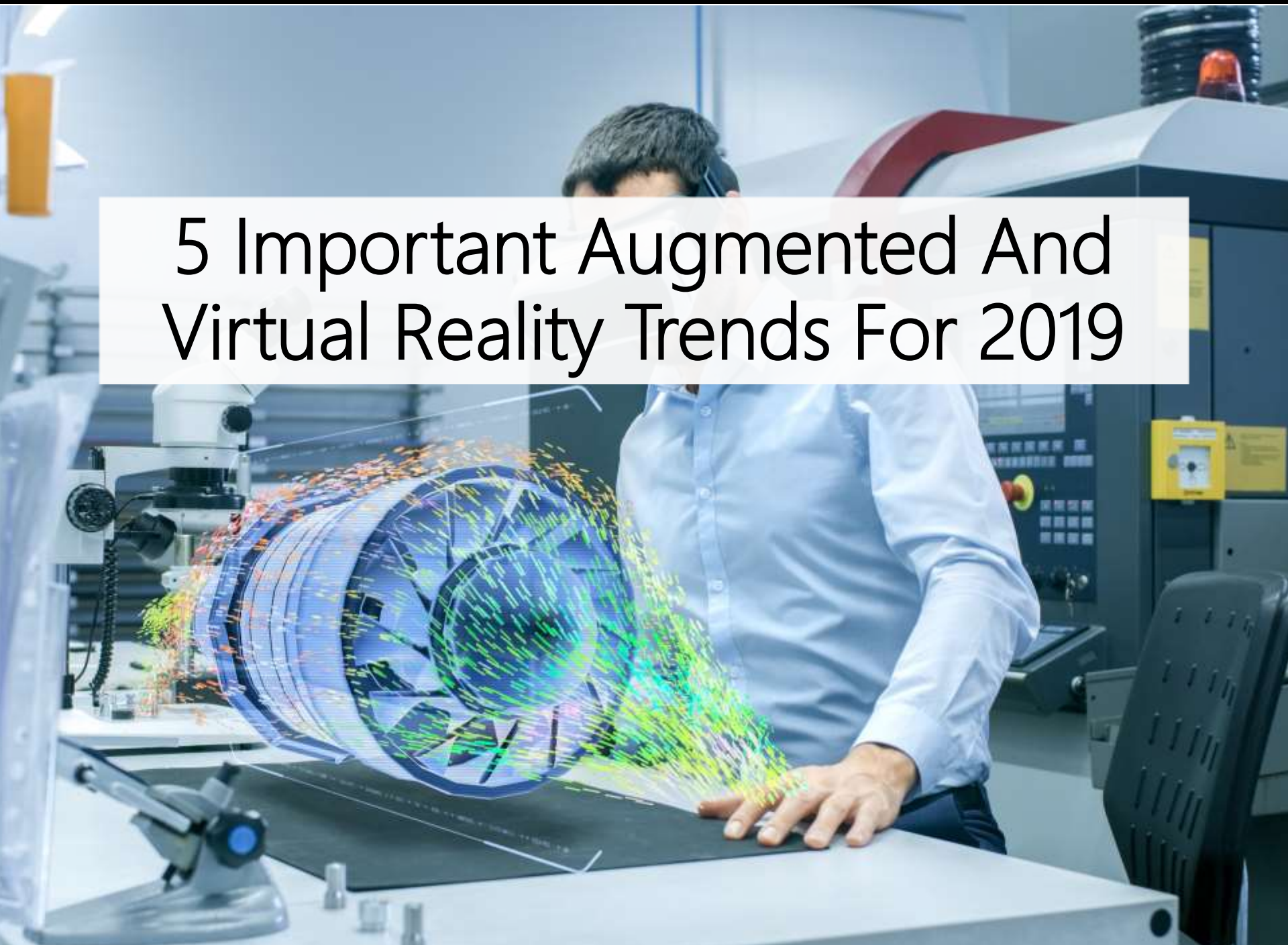


5 Important Augmented And Virtual Reality Trends For 2019

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Introduction

Alongside AI and automation, virtual reality (VR) and its closely related cousin augmented reality (AR) have been touted for several years now as technologies likely to have a profoundly transformative effect on the way we live and work.

Solutions which allowing humans to explore fully immersive computer-generated worlds (in VR), and overlay computer graphics onto our view of our immediate environment (AR) are both increasingly being adopted in both entertainment and industry.

Over the next year, both VR and AR applications will become increasingly sophisticated, as devices get more powerful and capable of creating higher quality visuals. Our understanding of how humans can usefully navigate and interact within virtual or augmented environments will also evolve, leading to the creation of more "natural" methods of interacting and exploring virtual space.

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AR and VR increasingly enhanced with AI

In a collision of two-letter abbreviations unlike anything that has come before it, AR and VR developers will increasingly build smart, cognitive functionality into their apps.

Computer vision – an AI (artificial intelligence) technology which allows computers to understand what they are "seeing" through cameras, is essential to the operation of AR, allowing objects in the user's field of vision to be identified and labeled. We can expect the machine learning algorithms that enable these features to become increasingly sophisticated and capable.

The Snapchat and Instagram filters we are used to, to, e.g. overlay bunny ears and cat whiskers on selfies, are a very consumer-facing application of AI tech combined with AR. Their popularity in these and various other applications of image enhancement functionality isn't likely to dwindle in 2019.

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AR and VR increasingly enhanced with AI

For more scientific use cases, there's Google's machine learning-enabled microscope to look forward to, which can highlight tissue which it suspects could be a cancerous tumor growth as a pathologist is looking at samples through the viewfinder.

VR is about putting people inside virtual environments and those environments – and their inhabitants – are likely to become increasingly intelligent over the next year. This is likely to include more voice control stemming from AI natural language processing, increasing immersion by reducing the reliance on icons and menus intruding into the virtual world. Gamers in VR will also face more challenging opponents as computer-controlled players will more effectively react and adapt to individual play styles.

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VR and AR will increasingly be used in training and teaching

Both technologies have obvious use cases in education. Virtual environments allow students to practice anything from construction to flight to surgery without the risks associated with real-world training. While augmented environments mean, information can be passed to the student in real time on objectives, hazards or best-practice.

This year Walmart announced that it is using 17,000 Oculus Go headsets to train its employees in skills ranging from compliance to customer service. In particular, training in the use of new technology is a focus for the retailer, with staff learning to use the new Pickup Tower automated vending units in virtual environments before they were deployed in stores.

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VR and AR will increasingly be used in training and teaching

Additionally, the US Army has announced a deal with Microsoft to use its HoloLens technology in military training, meaning soldiers will get real-time readings on their environment. Currently, this includes readouts to provide real-time metrics on soldier performance such as data about heart and breathing rates, but research objectives are to develop pathfinding, target acquisition and mission planning.

As VR and AR both continue to prove their worth at reducing risk and cost associated with training, it is likely we will see an increasingly rapid pace of adoption in industries involving work with expensive tools and equipment, or hazardous conditions, throughout 2019.

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Consumer Entertainment VR hits the mainstream

Ok, this one has been predicted for a couple of years now. VR adoption in homes has been steady since consumer headsets hit the market a couple of years ago, but hardware and application developers haven't quite hit the sweet spot yet when it comes to creating the VR "killer app."

But some significant developments are coming up that could mean 2019 is the year we start to see the real action here. Previous generations of VR headsets have been limited in one of two ways. Either by the user having to be tethered to a big, expensive computer to power the "experience," hence limiting our mobility and therefore the sense of immersion. Or by relying on relatively low-powered mobile tech to control stand-alone headsets, meaning graphics quality is limited – another immersion-breaker.

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Consumer Entertainment VR hits the mainstream

This year, stand-alone headsets incorporating powerful, dedicated computer technology will hit the shelves, from both Vive and Oculus. Confident that their users will now be unrestricted by cables or low-powered displays, VR developers will create more realistic and accurate simulations of our real world within their virtual worlds. This will mean more immersive entertainment experiences and an unprecedented level of realism within VR games.

As well as being mobile, the new generation of headsets will improve the technology powering the virtual experience, by including features such as eyeball-tracking and increased field-of-view. Again, this will help users feel they can interact and explore in more natural ways.

Of course, it isn't just the major players who are innovating – in a market like VR there's always room for an underdog to shake things up. Amazon lists over 200 different VR headsets available to buy, many of them being created by startups promising new features and functionality that could end up being game-changers.

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VR and AR environments becoming increasingly collaborative and social

Facebook's purchase of Oculus in 2016 showed that the social media giant believed virtual reality would become vital to the way we build shared online environments. Whether it's for virtual "conference calls" where participants can see and interact with each other, or socializing and relaxing with friends.

Pioneers such as Spatial are leading the way with AR tools for the boardroom and office, where users can see virtual whiteboards and pin boards, as well as collaboratively work on design documents overlaid on real-world objects.

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VR and AR environments becoming increasingly collaborative and social

This year, I am also expecting to see Facebook's VR Spaces platform, which allows users to meet and socialize in VR, move out of beta, and Tencent has announced that it is looking into adding VR to its WeChat mobile messaging system – the most widely used messenger app in the world.

Combined with the predicted increase in sales of VR and AR headsets, this could mean that 2019 is the year we experience meeting and interacting with realistic representations of our friends and family in VR, for the first time.

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AR increasingly finding its way into vehicles

Fully (level 5) autonomous cars may still be a few years away from becoming an everyday reality for most of us, but automobile manufacturers have plenty of other AI tech to dazzle us with in the meantime. Two of the most significant trends in new vehicles in 2019 will be voice assistants – with most major manufacturers implementing their takes on Alexa and Siri – and in-car AR.

Powered by machine learning, Nvidia's DriveAR platform uses a dashboard-mounted display overlaying graphics on camera footage from around the car, pointing out everything from hazards to historic landmarks along the way. Audi, Mercedes-Benz, Tesla, Toyota, and Volvo have all signed up to work with the technology.

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AR increasingly finding its way into vehicles

Alibaba-backed startup WayRay takes the route of projecting the AR data directly onto the car windshield, giving navigation prompts, right-of-way information, lane identification, and hazard detection.

In-car AR has the potential to improve safety – by allowing the driver to keep their eyes on the road as they read feedback that would previously have been given on a sat-nav or phone screen, as well as increase comfort and driver convenience. In a few years, it's likely we will wonder how we ever lived without it.

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Bernard Marr is an internationally best-selling author, popular keynote speaker, futurist, and a strategic business & technology advisor to governments and companies. He helps organisations improve their business performance, use data more intelligently, and understand the implications of new technologies such as artificial intelligence, big data, blockchains, and the Internet of Things.

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